

# Multidimensional Goodhart

How controlling for a proxy reshapes, rather than removes, its error

Working draft — Chapters 1–3

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# 1 Goodhart’s law, multidimensionally

## 1.1 The shape of the problem

Goodhart’s law is usually quoted in one of two forms. The popular slogan, which is actually Strathern’s compression [6]:

When a measure becomes a target, it ceases to be a good measure.

and Goodhart’s own statement [2]:

Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.

Both are stated about a **scalar** measure: a single number that was correlated with something we cared about, until we started optimising it. The thesis of this book is that the scalar framing hides most of the structure. Real goals — the goals of a person, a team, a company, a state, or a trained model — are **multidimensional**, and the proxies we use to steer toward them are multidimensional too, usually with a different (and smaller) set of dimensions. Once that is taken seriously, “the regularity collapses” is replaced by a more precise and more useful claim: as you add structure to a proxy in order to control for Goodhart in the dimensions you can see, the residual error does not vanish. It moves — into dimensions you cannot see, or into a form (strategic, adversarial, harder to summarise) that the tools that controlled the original error do not reach.

This is not, by itself, a theorem; it is a research programme. The job of these first chapters is to build the vocabulary precisely enough that the programme’s sub-claims become statable, and to prove the ones that are currently provable. The starting point is the variant taxonomy of Manheim and Garrabrant [4] — regressional, extremal, causal, and adversarial Goodhart — and Wentworth’s geometric reconstruction of the regressional case [7]; the contribution here is to make the **vector** structure (and the selection-vs-intervention split that organises that taxonomy) explicit.

## 1.2 A worked intuition: the hierarchy of proxies

Consider a company with leadership whose real objectives are something like *make a product customers want, make money, enjoy the work, gain prestige*, and a dozen other things, in some weighting that no written document captures. Call that vector of objectives  $G$ . Each quarter the leadership sets a proxy  $P$ : a handful of metrics — revenue, delivery dates, an NPS number. The teams that implement  $P$  have their own objectives, and the proxy they **actually** optimise,  $P_i$ , folds their incentives into the official one. Its dimensionality is at least that of the most complex implementer’s goals.

Now suppose leadership notices that Team 1 is “optimising too hard on the metrics” — the dashboard looks better than the product. They intervene. The correction they apply is *not* the gap between  $G$  and Team 1’s true output. It is the gap between the **felt proxy** by which leadership noticed the problem and Team 1’s measured performance. The error term is being filtered through several layers of proxy, each with its own residual.

Two things follow, and they recur throughout the book:

1. When a simpler agent (fewer goal dimensions) sets goals for a more complex agent, the proxy it can write down is necessarily missing dimensions of the complex agent’s behaviour — and therefore **cannot even score** the complex agent’s drift in those dimensions. If a human writes a proxy intended to align a system whose objectives have more dimensions than the human’s, the proxy contains error in dimensions the human’s  $G$  has no coordinate for. Call it incomputable noise: not noise the human is failing to measure well, but noise the human’s objective cannot be *compared against at all*.
2. Whether the multi-layer correction process converges depends on whether the layers’ perception errors are roughly independent. If each layer’s residual is its own, a dimension of error that one layer sees can be corrected even though no layer sees the whole picture. If the residuals are strongly correlated — a centrally-planned organisation, a single dominant metric — the system is fragile: every layer is blind in the same direction at once.

*Remark.* There is an ML reading of all this. Optimising an empirical scalar proxy selects for large residual generalisation error; that is regressional Goodhart. The vector version says the residual is not one scalar gap but a **subspace** of gaps, some of which are instrumented and some of which are not. Existing hyperparameter-optimisation and distribution-shift theory already implements pieces of this picture; we will point at the correspondences as they arise rather than develop them.

### 1.3 Setup: goals, proxies, and two kinds of gap

Fix a state space  $S$  — the set of configurations the world (or the organisation, or the model) can be in. A **true goal map** is a function  $G : S \rightarrow \mathbb{R}^m$ ; its  $m$  coordinates are substantively distinct things the principal cares about. A **proxy map** is  $P : S \rightarrow \mathbb{R}^k$ , the  $k$  things actually measured and acted on. The principal also has in mind an **intended model**  $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$  — “the proxy should be roughly  $\varphi$  of the goal” — and the discrepancy is the **residual**

$$\varepsilon(s) = P(s) - \varphi(G(s)).$$

This little decomposition already separates two distinct Goodhart channels:

- A **dimension gap**:  $\ker \varphi \neq \{0\}$ , so some directions in goal space are invisible to the proxy. No amount of measuring  $P$  accurately tells you anything about movement along  $\ker \varphi$ .
- An **observation gap**:  $\varepsilon \neq 0$ , so even the directions the proxy is meant to track are corrupted — by noise, by lag, or (Chapter 3) by deliberate manipulation.

**Toy example.** Let product quality be  $G = (\text{reliability}, \text{delight})$  and let the proxy be uptime. Then “delight” lies in  $\ker \varphi$  — the dimension gap. Noisy uptime instrumentation is part of  $\varepsilon$  — the observation gap. A research-evaluation example: citation count is **intended** to track research quality; “long-run conceptual fertility” may be in  $\ker \varphi$ , while bot citations and citation cartels are residual variation in proxy space.

*Remark.* A caution that will shape the notation. If the principal’s “true” objective is genuinely scalar, then  $\ker \varphi \neq \{0\}$  may be an artefact of having written that scalar goal in redundant coordinates. So dimension-gap claims should always be made **after** choosing coordinates whose components are substantively distinct goal variations, not arbitrary embeddings. We will keep  $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$  as the principal’s intended correspondence and consistently distinguish three subspaces:

- $\ker \varphi$ : goal variation invisible to the intended proxy;
- $\text{im } \varphi$ : proxy variation intended to track goal variation;
- residual proxy directions: variation of  $P$  not explained by  $\varphi(G)$ .

## 1.4 A first specialisation, and a roadmap

Throughout Chapter 2 we work the simplest non-trivial case:  $G(s) = X \in \mathbb{R}^m$  with  $X$  Gaussian, a single scalar proxy that is one coordinate (or one linear combination) of  $X$  plus noise, and **selection** — keep the states scoring above a threshold. This isolates the two classical “easy” Goodhart effects, **regressional** and **extremal** Goodhart [4], from the harder ones. It is the right first probe precisely because it is the regime where, as Chapter 2 shows, hidden harm turns out to be **bounded by quantities visible in the pre-selection distribution**. Chapter 3 then asks what changes when the principal’s policy does not merely re-select from a fixed population but changes the population itself — when agents **respond** — and shows that this is where the “deep Goodhart” story has teeth, and where the bound from Chapter 2 has no analogue.

A reader who wants the punchline before the construction: there is a clean dichotomy between **selection channels** (the policy reweights a fixed baseline distribution; all classical regressional/extremal Goodhart lives here; hidden drift  $\leq \delta \cdot \|s\|$  with every term a baseline functional) and **intervention channels** (the policy moves probability mass to where the baseline had none; causal and adversarial Goodhart live here; no baseline bound exists, and any bound must be imported from a model of what the responding agents can afford to do). Chapters 2 and 3 are those two halves.

## 2 Selection channels: when the principal only re-selects

### 2.1 The Gaussian threshold model

Let  $X = (X_1, \dots, X_m) \sim \mathcal{N}(0, \Sigma)$ . Take the proxy to be the first coordinate,  $P = X_1$ , with selection  $A_t = \{X_1 \geq t\}$ ; the **hidden** goal coordinates are  $H = (X_2, \dots, X_m)$ , so in this stripped-down picture  $\dim(\ker \varphi) = d = m - 1$ . (We have set the observation noise to zero on purpose: whatever effect appears here comes from the dimension gap, not from measurement error.)

For each hidden coordinate write  $\rho_j = \frac{\text{Cov}(X_j, X_1)}{\text{Var}(X_1)}$ . The Gaussian conditional-mean formula gives  $\mathbb{E}[X_j \mid X_1 = x] = \rho_j x$ , and hence

$$\mathbb{E}[X_j \mid A_t] = \rho_j \cdot \mathbb{E}[X_1 \mid X_1 \geq t].$$

If  $X_1 \sim \mathcal{N}(0, \sigma_1^2)$  then  $\mathbb{E}[X_1 \mid X_1 \geq t] = \sigma_1 \lambda(\alpha)$  with  $\alpha = \frac{t}{\sigma_1}$  and  $\lambda(\alpha) = \frac{\phi(\alpha)}{1 - \Phi(\alpha)}$  the inverse Mills ratio. So the hidden mean-shift vector is

$$\mathbb{E}[H \mid A_t] = \sigma_1 \lambda(\alpha) \cdot r, \quad r = (\rho_2, \dots, \rho_m).$$

#### Selection moves only what correlates with the proxy (in this model).

In the Gaussian threshold model, thresholding on  $X_1$  produces no expected drift in a hidden coordinate  $X_j$  exactly when  $\text{Cov}(X_j, X_1) = 0$ . *Toy example:* if a school rewards only test scores and student curiosity is statistically independent of test scores in the applicant pool, thresholding on scores does not change mean curiosity.

*Remark.* This is a **Gaussian** fact and must not be read as a general one — see Section 2.3. It is true here only because Gaussian conditional expectations are linear; for non-Gaussian variables, zero covariance does not imply a flat tail-conditional mean. Read correctly, it is a **sufficiency** statement: in the multivariate-Gaussian scalar-threshold model, the covariance ratios  $r$  are a complete summary of hidden mean drift at every threshold.

### 2.2 Does the harm scale with the number of hidden dimensions?

The motivating intuition wants “more hidden dimensions  $\Rightarrow$  more Goodhart”. The honest version is weaker.

If every hidden dimension has the **same** covariance ratio  $\rho_j = \rho$ , then

$$\|\mathbb{E}[H \mid A_t]\|_2 = \sqrt{d} \cdot |\rho| \cdot \sigma_1 \lambda(\alpha),$$

so the aggregate hidden mean drift grows like  $\sqrt{\dim(\ker \varphi)}$  at fixed per-dimension coupling. But “fixed per-dimension coupling” is a strong scaling assumption. If instead the **total** correlation budget is bounded,  $\sum_j \rho_j^2 \leq C$ , then  $\|\mathbb{E}[H \mid A_t]\|_2 \leq$

$\sigma_1 \lambda(\alpha) \sqrt{C}$ , which does not grow with  $d$  at all. Whether dimensional growth is real depends entirely on whether new dimensions bring new **independent** coupling to the proxy or merely subdivide a fixed amount of coupling into finer coordinates.

The clean first version of dimensional dependence is therefore **not** “more hidden dimensions automatically cause more Goodhart”. It is: under a per-dimension coupling model, threshold selection induces hidden drift whose **norm scales with the coupling-vector norm**  $\|r\|_2$ ; this becomes **dimensional** scaling only after a substantive assumption about how  $\|r\|_2$  grows with  $d$ .

*Remark.* Two negative results worth recording, because they kill the obvious shortcuts. (i) If hidden dimensions are independent of the proxy, thresholding leaves the hidden distribution unchanged — more hidden dimensions alone do nothing. (ii) The **signed** aggregate hidden error is the wrong target: positive and negative hidden correlations cancel in a signed sum even while the normed or squared hidden displacement grows. Use normed drift, squared loss, tail probability, or a domain-specific loss — never the raw signed sum. And for a squared-harm functional  $L(H) = \|H\|_2^2$ , selection can raise harm through **either** a mean shift or a change in conditional variance, so the mean-drift vector alone does not determine expected harm.

### 2.3 Covariance is not a general coupling primitive

Outside Gaussian linearity, covariance fails. Let  $P = Z$  with  $Z \sim \mathcal{N}(0, 1)$  and let the hidden coordinate be

$$H = Z^2 - 1.$$

Then  $\mathbb{E}[H] = 0$  and  $\text{Cov}(H, P) = \mathbb{E}[(Z^2 - 1)Z] = \mathbb{E}[Z^3] - \mathbb{E}[Z] = 0$ . But under threshold selection  $A_t = \{Z \geq t\}$ ,

$$\mathbb{E}[H \mid A_t] = \mathbb{E}[Z^2 - 1 \mid Z \geq t] = t\lambda(t) > 0 \quad \text{for } t > 0.$$

Zero covariance between a hidden coordinate and a scalar proxy does **not** imply zero hidden drift under threshold selection. *Toy example:* an academic evaluation score may be uncorrelated with intellectual conformity overall — because both very-low-score and very-high-score candidates are unusual — while selecting only the very highest scores still enriches for one particular kind of unusualness.

*Remark.* The example writes  $H$  as a deterministic function of  $P$ , which feels engineered; but that is exactly the point against covariance — nonlinear dependence can be invisible to covariance while completely determining tail behaviour. A less deterministic version  $H = Z^2 - 1 + \xi$  with independent mean-zero  $\xi$  keeps the same covariance and the same tail-mean shift while weakening the functional-dependence objection. The right takeaway is **not** that real systems generically have U-shaped hidden dependence; it is that covariance cannot be the universal coupling primitive.

The repair is to take the **response curve itself** as primitive. For a scalar proxy  $P$  and hidden vector  $H$ , define the **threshold response**

$$b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H],$$

whenever the conditional expectation exists. This describes the actual displacement caused by selection at pressure level  $t$ , with no linearity assumption. In the Gaussian scalar model it reduces to what we already have:  $b_{H(t)} = \sigma_1 \lambda\left(\frac{t}{\sigma_1}\right)r$ , so  $r$  is a sufficient statistic for **all** threshold responses in that restricted model — but only there.

*Remark.*  $b_{H(t)}$  is a **threshold-selection** primitive, not the final one. It says nothing about smoother optimisation policies — funding everyone above a cutoff is one thing; funding probabilistically with odds increasing in score is another. The next section generalises.

## 2.4 Weighted selection response

Let  $(S, \mathcal{F}, \mu)$  be the baseline probability space,  $H : S \rightarrow \mathbb{R}^d$  the hidden coordinates,  $P : S \rightarrow \mathbb{R}$  the scalar proxy. A **selection policy** is a nonnegative weight  $W_\theta : S \rightarrow [0, \infty)$  with  $0 < \mathbb{E}_{\mu[W_\theta]} < \infty$ . Define the **selected expectation** and the **hidden response**

$$\mathbb{E}_{\theta[F]} = \frac{\mathbb{E}_{\mu[F W_\theta]}}{\mathbb{E}_{\mu[W_\theta]}}, \quad B_{H(\theta)} = \mathbb{E}_{\theta[H]} - \mathbb{E}_{\mu[H]}.$$

Hard thresholding is the special case  $W_t = \mathbb{1}\{P \geq t\}$  (recovering  $b_{H(t)}$ ). Soft optimisation is  $W_\beta = \exp(\beta P)$  — Boltzmann selection — whenever  $\mathbb{E}[\exp(\beta P)]$  is finite. Probabilistic funding with score-increasing odds, replicator-style repeated reweighting by performance, top- $q$ -fraction selection (thresholding at an endogenous quantile): all are weight functions.

*Remark.* Two caveats on the soft-optimisation case. First, if  $P$  is heavy-tailed,  $\mathbb{E}[\exp(\beta P)]$  may be infinite for positive  $\beta$  — the Boltzmann path may not exist. This is not a technicality; Goodhart is often precisely about extreme tails. For heavy tails, bounded weights or quantile selection are safer models. Second, not every control process is a reweighting of a fixed baseline at all — interventions can change the state-generating mechanism. That is Chapter 3.

### 2.4.1 Covariance as a local velocity

For Boltzmann selection,  $\mathbb{E}_{\beta[H]} = \frac{\mathbb{E}[H \exp(\beta P)]}{\mathbb{E}[\exp(\beta P)]}$ , and assuming enough integrability to differentiate under the expectation,

$$\frac{d}{d\beta} \mathbb{E}_{\beta[H]} = \mathbb{E}_{\beta[HP]} - \mathbb{E}_{\beta[H]} \mathbb{E}_{\beta[P]} = \text{Cov}_{\beta(H, P)}.$$

Covariance is best read as the **local velocity** of hidden drift under infinitesimal soft optimisation, evaluated under the **current** selected distribution — not as a global finite-pressure summary. *Toy example:* at low bonus pressure, the rate at which burnout changes with sales incentives is the current covariance between



burnout and sales; after employees adapt or the selected population shifts, the covariance must be recomputed under the new weighted distribution.

That local velocity does not pin down finite movement. Take again  $P = Z \sim \mathcal{N}(0, 1)$ ,  $H = Z^2 - 1$ . Boltzmann tilting by  $\exp(\beta Z)$  gives  $Z_\beta \sim \mathcal{N}(\beta, 1)$ , so

$$\mathbb{E}_{\beta[H]} = \mathbb{E}[Z_\beta^2 - 1] = \beta^2.$$

The covariance at  $\beta = 0$  is zero — matching Section 2.3 — yet finite pressure gives strictly positive hidden drift for every  $\beta \neq 0$ . Baseline covariance alone is insufficient even for **finite** soft optimisation; one needs the covariance field  $\text{Cov}_{\beta(H,P)}$  along the whole tilted path.

**The hierarchy of selection primitives.** covariance (infinitesimal Boltzmann velocity)  $\subset$  threshold response  $b_{H(t)}$  (hard cutoffs)  $\subset$  weighted response  $B_{H(\theta)}$  (generic non-causal selection). All three are functionals of the baseline  $\mu$  alone. Causal and adversarial Goodhart need a further layer in which  $\mu$  itself changes with the principal’s policy — the subject of Chapter 3.

## 2.5 The selection-channel drift bound

This is the structural payoff of staying inside the reweighting picture. With  $W_\theta$  a selection policy, write the likelihood ratio  $L_\theta = \frac{W_\theta}{\mathbb{E}_\mu[W_\theta]}$ , so the selected law is  $\mu_\theta = L_\theta \mu$ . By Cauchy–Schwarz in  $L^2(\mu)$ , for each hidden coordinate  $H_i$ ,

$$|B_{H_i}(\theta)| = \left| \mathbb{E}_\mu[(L_\theta - 1)(H_i - \mathbb{E}_\mu H_i)] \right| \leq \|L_\theta - 1\|_{L^2(\mu)} \cdot \|H_i - \mathbb{E}_\mu H_i\|_{L^2(\mu)}.$$

So, writing  $\delta := \|L_\theta - 1\|_{L^2(\mu)}$  — a **reweighting budget**, with  $\delta^2 = \chi^2(\mu_\theta \parallel \mu)$  the chi-square divergence — and  $s_i := \text{sd}_{\mu(H_i)}$ ,

$$|B_{H_i}(\theta)| \leq \delta \cdot s_i, \quad \|B_{H(\theta)}\|_2 \leq \delta \cdot \|s\|_2.$$

Under a selection channel, hidden Goodhart drift is controlled by two things that **are** visible in the baseline distribution: how hard you reweight ( $\delta$ ) and how variable the hidden coordinates are ( $s$ ). Bounded reweighting budget plus bounded hidden variance  $\Rightarrow$  bounded hidden drift — full stop. The  $\sqrt{d}$  growth from Chapter 2’s per-dimension model is exactly the  $\|s\|_2$  term, and that term is the **only** way the number of dimensions enters.

*Remark.* This bound is one line of Cauchy–Schwarz, and it should not be inflated into a headline theorem. It is a worst-case envelope, not a prediction (the actual drift is the inner product, not the product of norms), and it is vacuous when  $\delta$  is large — extreme selection such as “top  $1/n$ ” can have  $\delta$  growing with sample size. Its job is purely structural: **every term on the right is a  $\mu$ -functional**. The content of Chapter 3 is the contrast — that intervention channels admit no such bound, not the inequality itself.

# 3 Intervention channels: when agents respond

## 3.1 Why selection is not enough

A selection policy can only move probability mass that already exists toward high- $P$  regions: the selected law  $\mu_\theta = L_\theta \mu$  is absolutely continuous with respect to  $\mu$ , so any event with baseline probability zero keeps probability zero. Causal and adversarial Goodhart violate exactly this. Announcing a metric changes how agents behave; probability mass appears in parts of state space the baseline never visited. Teaching to the test, metric-specific optimisation, outright fabrication — none of these are reweightings of last year’s population. They **transport** mass to new places.

**The sharp boundary.** The selection/intervention distinction is substantive exactly when agents can **move in state space** — change their  $(P, H)$  at a fixed underlying type — not merely toggle their own inclusion. If agents only choose whether to **participate** (apply, submit, enter the pool) but cannot alter their measured features, then the post-policy law is just  $\mu$  restricted to the participating set and renormalised — a selection channel after all. Genuine causal Goodhart lives in features that are **cheap to change without changing the thing they were supposed to proxy**.

## 3.2 Response channels: the top-level object

Let  $(S, \mathcal{F})$  be a measurable state space,  $\mu_0$  the baseline law,  $H : S \rightarrow \mathbb{R}^d$  the hidden goal coordinates,  $P : S \rightarrow \mathbb{R}$  the scalar proxy, and  $\varphi$  the principal’s intended model  $P \approx \varphi(G)$ . A **response channel** is a map

$$\mathcal{R} : \Theta \rightarrow \mathcal{P}(S), \quad \theta \mapsto \mu_\theta, \quad \mu_{\theta_0} = \mu_0,$$

for some null policy  $\theta_0$ . Hidden drift along the channel is  $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu_0}[H]$ , exactly as before.

**Definition.**  $\mathcal{R}$  is a **selection channel** if  $\mu_\theta \ll \mu_0$  for all  $\theta$  — then  $L_\theta := d\mu_\theta/d\mu_0$  exists and  $\mu_\theta$  is  $\mu_0$  reweighted by  $L_\theta$ . Otherwise  $\mathcal{R}$  is an **intervention channel**:  $\mu_\theta$  may be mutually singular with  $\mu_0$ .

The entire apparatus of Chapter 2 — covariance, threshold response, weighted response — is the selection-channel case, with  $L_\theta = \frac{W_\theta}{\mathbb{E}_{\mu_0}}[W_\theta]$ . In Manheim and Garrabrant’s taxonomy [4], **causal** Goodhart is an intervention channel in which the policy structurally breaks  $P \approx \varphi(G)$ , and **adversarial** Goodhart is an intervention channel in which  $\theta$  is chosen worst-case for the principal. The ML instances of this regime are **strategic classification** [3] — agents manipulate features in response to a published classifier — and **performative prediction** [5] — the act of deploying a predictor changes the distribution it is predicting. The drift bound

of Chapter 2 has no analogue here: there is no  $L_\theta$ , so nothing plays the role of  $\delta$ . Any bound on intervention drift must be **imported from the agent side** — a feasibility set or cost function describing what re-arrangements of mass agents can afford. The rest of the chapter computes such a bound in toy models and reads off the controlling quantity.

### 3.3 A linear–Gaussian Stackelberg gaming model

A continuum of agents; an agent’s type is its true quality  $Q \sim \mathcal{N}(0, \sigma_Q^2)$ , and the true scalar goal is  $G = Q$ . There is a hidden coordinate  $H$  = “gaming externality”: effort spent corrupting the proxy rather than improving  $Q$  — teaching to the test, metric-specific optimisation, Campbell’s-law distortion [1]. The principal believes  $P \approx Q$ .

**Baseline channel  $\mu_0$**  (metric not announced, not gameable): the agent does nothing special,  $P_0 = Q + \eta$  with  $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$  independent, and  $H_0 \equiv 0$ . So  $\mu_0$  is supported on the plane  $\{H = 0\}$  in  $(Q, P, H)$ -space.

**Intervention channel** (metric announced, agents best-respond; Stackelberg, principal commits first): the principal picks a threshold policy  $A_t = \{P \geq t\}$ ; selection is worth  $V > 0$  to an agent. An agent of type  $Q$  chooses gaming effort  $a \geq 0$ , getting  $P = Q + a + \eta$  at private cost  $c(a) = a^2/(2\kappa)$  ( $\kappa > 0$  measures the ease of gaming) and incurring hidden externality  $H = a$ . The agent maximises  $V \cdot \Pr(Q + a + \eta \geq t) - a^2/(2\kappa)$ .

#### 3.3.1 The noiseless cut

Set  $\sigma_\eta = 0$ , so selection is  $Q + a \geq t$ . An agent with  $Q \geq t$  sets  $a = 0$ . An agent with  $Q < t$  either games to exactly  $a = t - Q$  (cost  $(t - Q)^2/(2\kappa)$ , payoff  $V$ ) or sets  $a = 0$  (payoff 0). Gaming is worth it iff  $(t - Q)^2/(2\kappa) \leq V$ , i.e.

$$t - Q \leq \Delta, \quad \Delta := \sqrt{2\kappa V}.$$

So the best response is  $a^*(Q) = (t - Q) \cdot \mathbb{1}\{t - \Delta \leq Q < t\}$ , and the selected set is  $\{Q \geq t - \Delta\}$ . Compared to the **selection** counterfactual (agents cannot game; the principal thresholds  $P = Q$  at the same nominal  $t$ ; selected set  $\{Q \geq t\}$ ;  $H \equiv 0$ ):

1. **The metric lies, by up to  $\Delta$ .** The principal believes selection  $\Rightarrow P \geq t \Rightarrow$  (under  $\varphi$ )  $Q \geq t$ . Actually  $Q$  among the selected ranges down to  $t - \Delta$ . The proxy-to-goal gap  $\varepsilon = P - \varphi(G) = a$  is no longer noise — it is a deliberate wedge of size up to  $\Delta$ .
2. **Hidden harm appears at order  $\Delta$ .**  $\mathbb{E}[H \mid \text{selected}] = \mathbb{E}[(t - Q)\mathbb{1}\{t - \Delta \leq Q < t\}] / \Pr(Q \geq t - \Delta) > 0$ , monotone increasing in  $\Delta = \sqrt{2\kappa V}$ . In the selection regime  $H$  is **identically zero** in this model — consistent with Chapter 2’s bound, since  $s_H = 0 \Rightarrow B_H = 0$ . The harm is created entirely by the intervention.
3. **The controlling quantity is not a  $\mu_0$ -functional.**  $\Delta$  depends on  $\kappa$  (technological ease of gaming) and  $V$  (the stakes), neither of which is visible in the baseline  $(Q, P_0, H_0)$  law. This is the promised contrast: intervention drift is bounded by the agents’ cost–benefit ratio, not by any reweighting budget.

4. **Why this is not a selection channel.**  $\mu_0$  lives on  $\{H = 0\}$ ; the post-intervention law puts positive mass on  $\{H > 0\}$  (the gamers — a set of types of positive measure). The two laws are mutually singular in the  $(Q, H)$ -marginal: there is no  $L_\theta$ . The state was **transported**, not reweighted.

In a Stackelberg gaming model with quadratic gaming cost  $a^2/(2\kappa)$  and selection value  $V$ , the metric’s worst-case bias and the induced hidden harm both scale with  $\Delta = \sqrt{2\kappa V}$  — the square root of (ease of gaming  $\times$  stakes) — and this regime is invisible to, and unbounded by, the baseline distribution. *Toy example:* doubling the funding tied to a test score multiplies the gaming wedge by  $\sqrt{2}$ ; halving the cost of test-prep drilling does the same.

*Remark.* The quadratic cost is a modelling choice. A cost with a hard cap  $a \leq a_{\max}$  bounds gaming at  $a_{\max}$  regardless of  $V$ ; a cost that is cheaper at the margin for high- $Q$  agents skews **who** games. So  $\Delta = \sqrt{2\kappa V}$  is the quadratic-cost signature, not a universal law — but the qualitative point (a positive-measure set of agents leaves the  $H = 0$  locus; drift is set by agent economics, not by  $\mu_0$ ) is robust to the cost’s shape. And if  $V$  is itself endogenous — selection is valuable only if the metric is trusted, and trust erodes as gaming is observed — there is a feedback loop not modelled here. Section 3.5 flags it.

### 3.3.2 The noisy refinement (sketch)

With Gaussian proxy noise,  $\Pr(\text{selected} \mid Q, a) = 1 - \Phi((t - Q - a)/\sigma_\eta)$  is smooth in  $a$ , so the best response is interior everywhere:

$$a^*(Q) = (\kappa V / \sigma_\eta) \cdot \phi((t - Q - a^*(Q))/\sigma_\eta)$$

(implicit; the right side is the marginal selection-probability gain). Every agent games a little:  $a^*(Q) > 0$  for all  $Q$ , peaking near  $Q \approx t - a^*$  and decaying in  $|Q - (t - a^*)|$ . Hidden harm  $\mathbb{E}[H \mid \text{selected}] = \mathbb{E}[a^*(Q) \mid \text{selected}] > 0$  for every threshold. The hard-threshold version is the  $\sigma_\eta \rightarrow 0$  limit, where gaming concentrates on the band  $[t - \Delta, t)$ . The lesson: noise **spreads** gaming across the whole population rather than removing it.

## 3.4 Multidimensional gaming: does adding a measured dimension help?

Now the dimensional thread of Chapter 1 rejoins the picture, inside the intervention regime. The motivating “deep Goodhart” move is: the principal, fighting scalar Goodhart, **adds proxy dimensions**. When agents game, does adding a measured dimension redistribute harm, conserve it, shrink it, or grow it? The answer depends critically — and instructively — on how the metric **aggregates** its dimensions, so the aggregation rule must be a visible parameter, not a hidden default.

### 3.4.1 The additive model

Take  $k$  gaming channels and, in the base case, true quality  $Q = 0$  for everyone (pure gaming). An agent allocates effort  $a = (a_1, \dots, a_k) \geq 0$ ; channel  $j$  costs  $a_j^2/(2\kappa_j)$  and produces hidden harm  $H_j = a_j$  (all gaming equally wasteful,  $H = \sum_j a_j$ ). The principal **measures** a set  $M \subseteq \{1, \dots, k\}$  and scores additively,  $\text{score} = \sum_{j \in M} a_j$

(unit weights for now); selection (worth  $V$ ) iff score  $\geq t$ . An agent never games an unmeasured channel.

The agent's problem is  $\min_{a_j \geq 0, j \in M} \sum_{j \in M} a_j^2 / (2\kappa_j)$  subject to  $\sum_{j \in M} a_j \geq t$ , then game iff that minimum cost is  $\leq V$ . At the optimum the constraint binds and  $a_j / \kappa_j = \lambda$  for all  $j \in M$  (a water-filling allocation), so with  $K_M := \sum_{j \in M} \kappa_j$ ,

$$\lambda = t / K_M, \quad a_j = t \kappa_j / K_M, \quad \text{min cost} = \lambda^2 K_M / 2 = t^2 / (2K_M).$$

Hence **gaming occurs iff  $K_M \geq t^2 / (2V)$**  (call the right side  $K_{\min}$ ), and when it occurs, the per-agent hidden harm for this fixed pure-gaming target is  $H = \sum_{j \in M} a_j = \lambda K_M = t$  — **independent of  $M$** . What  $M$  controls is (i) **whether**  $K_M$  clears  $K_{\min}$ , and (ii) **how** the fixed harm  $t$  is split across the measured channels — proportional to  $\kappa_j$ .

**Conservation under re-routing (narrow form).** With a unit-weight additive metric and gaming that is equally wasteful per unit of score, the principal cannot reduce per-agent harm for a fixed pure-gaming score deficit by changing **which** channels it measures, as long as the measured set keeps enough aggregate gaming capacity ( $K_M \geq K_{\min}$ ): the harm is pinned at  $H = t$  and merely re-routes. Closing a gamed channel just spreads  $t$  over the others. *Toy example:* a hospital ranked on readmission rates clamps down on coding of “readmission” — the gaming reappears as patient selection, discharge timing, observation-status reclassification; the per-case distortion needed to clear the cutoff is unchanged.

**Adding a gameable measured dimension backfires.** Expanding  $M$  strictly increases  $K_M$ , which **lowers** the gaming cost  $t^2 / (2K_M)$ , which weakly **enlarges** the population that finds gaming worthwhile. With quality heterogeneity  $Q \sim \mathcal{N}(0, \sigma^2)$  — an agent needs score  $\geq t - Q$ , at cost  $(t - Q)^2 / (2K_M)$ , and games iff  $t - Q \leq \sqrt{2K_M V}$  — the gaming-eligibility cutoff  $Q \geq t - \sqrt{2K_M V}$  moves **down** as  $K_M$  grows. More agents game; the fraction of selected agents who are pure gamers rises. *Toy example:* a university worried that “publication count” is gamed adds “grant income” and “media mentions” to the scorecard; each is independently inflatable, so the cheapest path to any target score is now cheaper, and more faculty shift from research to portfolio-padding.

**The effective levers are aggregate, not allocative.** In this model the principal reduces gaming harm only by (a) shrinking aggregate gaming capacity  $K_M$  below  $K_{\min}$  — narrowing the measured set to channels that are individually hard to game (small  $\kappa_j$ ), or hardening channels; (b) raising the bar  $t$  (but this raises  $K_{\min}$  too; the net effect on the  $K_M \geq t^2 / (2V)$  test depends on whether real signal scales with  $t$ ); (c) cutting the prize  $V$ ; or (d) abandoning additive aggregation. Shuffling attention between channels of equal hardness does nothing. *Toy example:* anti-cheating in exams works by making the exam itself hard to game — proctoring, item rotation — not by adding more graded components.

### 3.4.2 A conjunctive metric flips the sign

Suppose instead that passing requires  $a_j \geq t$  for **every**  $j \in M$  — the principal demands the agent clear a bar on all measured dimensions. The cost-minimal way to pass is  $a_j = t$  for all  $j \in M$ , at cost  $\sum_{j \in M} t^2/(2\kappa_j)$ , and total harm  $H = \sum_{j \in M} t = t|M|$ . Now harm grows **linearly in the number of measured dimensions**: adding a dimension unambiguously increases per-gamer gaming harm — though it also raises the cost of clearing the metric, so fewer agents do, a genuine trade-off (harm-per-gamer up, gamer-count down).

Gaming harm’s dependence on the **number** of measured proxy dimensions is governed by the aggregation rule. Unit-weight compensatory/additive metrics conserve fixed-deficit per-agent harm ( $H = t$ , re-routing only) — when measured channels are equally harmful per score unit; conjunctive/min metrics **multiply** per-gamer harm ( $H = t|M|$ ). Real scorecards are usually compensatory (weighted sums of KPIs), which is the regime where “just add another metric” can backfire by cheapening the cheapest gaming path and expanding the gaming population. The selection-regime  $\sqrt{d}$ -type scaling from Chapter 2 and these intervention-regime flat/linear behaviours are **different phenomena** and should not be conflated.

*Remark.* Scope, made explicit. The additive claims above assume (1) unit-weight additive aggregation, (2) all gaming equally wasteful per unit score, (3) a fixed pure-gaming target deficit, and (4) the binding-constraint deterministic regime. Relax (2) — let channel  $j$  also contribute  $\gamma_j \in [0, 1]$  to the true goal, so  $H = \sum_j (1 - \gamma_j) a_j$  — and then  $H = \sum_{j \in M} (1 - \gamma_j) t \kappa_j / K_M$ , which the principal **can** shrink by steering effort onto high- $\gamma$  channels (measure the ones where “gaming” is half-real). So conservation is exactly the equally-wasteful idealisation; in practice the principal’s job is partly to pick proxies whose cheapest inflation is also partially the real thing. Relax (3) with quality heterogeneity and endogenous participation: aggregate population harm rises with  $K_M$  because more agents enter the gaming band. Relax (4) with proxy noise: agents overshoot for a safety margin, so per-gamer  $H$  rises slightly above the deterministic deficit, but the re-routing logic still applies. The clean additive/conjunctive dichotomy is the two extremes; real metrics live on the interpolation, and **where** on it is itself a design choice with predictable consequences.

### 3.4.3 The exchange-rate condition

The conservation claim above is the unit-weight special case of something sharper. Let the score be  $\sum_{j \in M} w_j a_j$ , costs  $\sum_j a_j^2/(2\kappa_j)$ , and hidden harm  $H = \sum_j h_j a_j$ . The cost-minimal allocation for a score deficit  $d$  is  $a_j = d \kappa_j w_j / W_M$  with  $W_M = \sum_{i \in M} \kappa_i w_i^2$ , and its harm is

$$H_M(d) = d \cdot \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

This is **not** invariant in  $M$ . (Two equally easy channels,  $\kappa_1 = \kappa_2 = 1$ , equal physical harm  $h_1 = h_2 = 1$ : measuring only channel 1 with  $w_1 = 2$  gives  $H = d/2$ ; measuring

only channel 2 with  $w_2 = 1$  gives  $H = d$ ; measuring both gives  $H = 3d/5$ .) Re-routing can raise **or** lower harm, depending on the score weights.

**Conservation, correctly stated.** For a fixed score deficit  $d$ , quadratic separable costs, and additive score  $\sum w_j a_j$ , per-agent harm is conserved under re-routing **if and only if** social harm is proportional to score contribution on every available channel —  $h_j = cw_j$  on the active measured set. Then  $H_M(d) = cd$ , independent of  $M$ ; otherwise  $H_M(d)$  depends on  $M$  through the cost-weighted average above. *Toy example:* if every point of score inflation is equally socially wasteful no matter which KPI supplies it, re-routing conserves harm; if grant-padding produces less waste per score point than citation-padding, moving weight toward grant-padding genuinely reduces harm.

*Remark.* This is the same structure as cost-benefit-weighted strategic classification [3], and as isoquant choice in production theory: agents pick the cheapest input bundle for a target output, while social damage is a **different** linear functional of the bundle. Conservation appears only when target output and social damage use the same exchange rates. One residual worry for later: real score weights are sometimes partly arbitrary normalisation choices, so future work must distinguish harmless unit changes from substantive incentive exchange rates.

### 3.5 What we have, and what is open

The two-chapter arc gives a clean dichotomy. **Selection channels** — the policy reweights a fixed baseline — contain all of regressional and extremal Goodhart; hidden drift is bounded,  $\|B_{H(\theta)}\|_2 \leq \delta \cdot \|s\|_2$ , with every term a baseline functional, and the number of dimensions enters only through  $\|s\|_2$ . **Intervention channels** — the policy transports mass to where the baseline had none — contain causal and adversarial Goodhart; there is no baseline bound, and the controlling quantity (e.g.  $\Delta = \sqrt{2\kappa V}$  in quadratic Stackelberg gaming) is exogenous to  $\mu_0$ , set by the responding agents’ cost geometry. In the multidimensional intervention regime, “fight Goodhart by measuring more dimensions” has a precise predicted failure mode: under a compensatory rule it conserves fixed-deficit per-agent harm only when channels are equally harmful per score unit, and it lowers the cheapest gaming cost, recruiting more gamers — a population-level backfire; under a conjunctive rule it multiplies harm by the number of bars. The principal’s real levers are aggregate (shrink total gaming capacity, harden channels, raise the bar relative to real signal, cut the prize) or structural (pick low-harm-per-score proxies whose cheapest inflation is partly real; change weights; or go conjunctive and accept harm scaling with the number of bars).

Two appendices catalogue what is not yet settled. Appendix A lists the questions **currently being worked on** — they have partial answers in toy models, but not yet at the level of polish the chapters above aim for. Appendix B lists questions that **surfaced during this work and are deliberately parked** — worth recording so they are not rediscovered from scratch, but not on the critical path.

## 4 Appendix A — Currently in progress

These are the live research questions feeding the chapters above. Each has a toy-model partial answer in the working notes; what is missing is generality, the right level of abstraction, or an adversarial pass that the chapter treatment would require.

**In progress. An intervention bound for general convex gaming costs.**

The selection-channel bound is  $\|B_H\| \leq \delta \cdot \|s\|$  (Chapter 2). Is there an intervention analogue that factors through the agents’ cost geometry — a “gaming budget” — rather than through any divergence from  $\mu_0$ ? The quadratic case gives  $\Delta = \sqrt{2\kappa V}$ . Conjecture: a version holds for general convex gaming costs, with the bound governed by the convex conjugate of the cost. Not yet stated or proved.

**In progress. The exact selection-class condition.**

Agents who can only toggle their own inclusion stay inside the selection class; agents who can move  $(P, H)$  at fixed type do not. Is “cannot change  $(P, H)$  at fixed type” the precise boundary, or are there genuine intermediate cases — e.g. agents can move  $P$  but  $H$  is pinned, or agents can move within a sub-manifold? A clean classification theorem is wanted.

**In progress. Adaptive hardening dynamics.**

A principal that each period hardens whichever measured channel is currently most-gamed (lowers its  $\kappa_j$ ) drives  $K_M$  down over time. Does this converge to no-gaming ( $K_M < t^2/(2V)$ ), and how fast? Is “harden the active channel” optimal among principal policies, or does committing to a narrow, hard-to-game  $M$  from the start dominate? Currently only a heuristic argument that attrition eventually wins.

**In progress. The measurement frontier.**

The principal must measure enough channels to recover real signal about  $G$ , but every gameable channel measured enlarges the attack surface  $K_M$ . Characterise the frontier between “informative enough” and “small enough attack surface”. Is it ever empty — i.e. are there goal/proxy geometries where no admissible measured set is both informative and safe?

**In progress. Sub- vs. super-modularity of  $H(M)$ .**

In the measured set  $M$  the unit-weight equal-harm additive case is modular-trivial ( $H = t$  on the gaming region, with a discontinuity at the  $K_M = K_{\min}$  boundary) and conjunctive is modular ( $H = t|M|$ ). Weighted aggregation and population entry may be



genuinely sub- or super-modular — which would say something about whether greedy principal policies (add/remove one channel at a time) are sane. Not yet computed.

**In progress. Per-agent vs. population welfare.** For heterogeneous quality  $Q$ , aggregate gaming harm is  $\mathbb{E}[(t - Q)\mathbb{1}\{0 < t - Q \leq \sqrt{2K_M V}\}]$  in the unit-weight additive model. How does this change under proxy noise, nonlinear costs, and endogenous  $V$  — and which aggregate (per-gamer, population total, population mean, tail) is the right welfare object for the framework? The chapters above currently slide between these; the book version should not.

**In progress. Endogenous stakes / performative fixed points.** Make  $V$  endogenous: selection is valuable only if the metric is trusted, and trust decays as gaming is observed. Does the principal–agent game have a performative-stable fixed point, and does hidden harm persist at it? Sketched as a Stackelberg toy with a trust state variable; the fixed-point analysis is not done.

## 5 Appendix B — Future open questions (not currently pursued)

These surfaced during the work and are recorded for later. They are not blocking anything in Chapters 1–3, and most need a substantial new piece of machinery.

**Future question. Spectral / basis decomposition of the error.** Three decompositions, each carrying different information, should be kept distinct: (i) the SVD of  $\varphi$  — right singular vectors are the  $G$ -directions  $\varphi$  distinguishes; small singular values mark structural blind spots, decomposing the **dimension gap**; (ii) the eigendecomposition of  $\text{Cov}(\varepsilon)$  — principal directions of measurement noise, decomposing the **observation gap**; (iii) the selection-induced shift of  $G$  — compare the law of  $G(s)$  before and after restricting to the selected set, the most diagnostic of actual Goodhart and the least obvious to compute. The bad-case corner is the **conjunction**: a  $G$ -direction where  $\varphi$  has a small singular value (low signal),  $\text{Cov}(\varepsilon)$  has mass (high noise), and the selection shift is large.

**Future question. Value-weighted Goodhart susceptibility.** PCA/SVD is value-neutral; Goodhart is value-laden. A huge principal component orthogonal to the value structure of  $G$  is harmless; a tiny one aligned with a load-bearing  $G$ -dimension can dominate harm. Proposed scalar per  $G$ -direction: (value importance)  $\times$  (1/ singular value of  $\varphi$ )  $\times$  (noise variance)  $\times$  (selection shift), under a value-weighted inner product on  $G$ -space. **Open**: whether the value-weighting requirement makes this a non-starter in practice, or whether reasonable proxies for value importance (the principal’s true utility gradient, or a proxy for it) recover useful structure. **Also open**: a strategic system can shape  $\varepsilon$  to lie in low-variance components of the historical observation while remaining maximally goal-harmful, so PCA-based detection is geometrically blind to value-aware adversaries — does the value-weighted version close that hole?

**Future question. Pareto selection rather than scalarisation.** Chapter 2 scalarises (a weighted score, then a threshold). Pareto-frontier selection preserves diversity across measured dimensions in a way scalar thresholding destroys. When does selecting the Pareto-undominated set compress the selected distribution enough to break identifiability of the **unmeasured** dimensions — and when does it not? This is a different selection operator and needs its own response theory.

**Future question. The  $\varphi$ -as-partial-observation alternative.** The chapters fix  $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$  as the principal’s intended correspondence. An alternative is to represent the proxy as a partial observation of  $G$  plus independent artifacts,

which changes what “ $\ker \varphi$ ” means and may interact better with partial-hypothesis formalisms. Worth a comparison; not done.

**Future question. Incomputable noise and agents more complex than their principals.** If a human writes a proxy intended to align a system whose objectives have **more** dimensions than the human’s, the proxy contains error in directions the human’s  $G$  has no coordinate for — error that cannot be scored, only suffered. Can partial hypotheses (in the style of infra-Bayesianism) recover any handle on this? And: is there any reason to expect a more capable system to have **fewer** goal dimensions than a human, rather than more? Both are wide open and somewhat speculative.

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